DUKE AIPI 590 · ALTERNATIVE DATA · FINAL

Climate Impacts on Futures Pricing



Measuring corn supply risk
and testing public weather signals

Haydn Stucker · November 18, 2026

The problem and the people it affects

Weather shocks create supply uncertainty for agricultural markets

Problem	Can public weather data measure crop shortfalls and add information beyond futures prices?
Target users	Commodity analysts first; corn buyers, feed mills, ethanol plants and lenders for the supply-risk result.
Why it matters	A shortfall can leave buyers under-provisioned. A false trading signal can turn a plausible story into losses.
Intended impact	Estimate physical exposure with uncertainty and test whether the same data supports market timing.

The final research product

A reproducible supply-risk package with a fixed model specification and model card

AT MIDSEMESTER	ADDED FOR THE FINAL
Leave-one-year-out yield estimates	Chronological forecasts against a trend baseline
Point estimates of crop damage	Intervals, coverage and failure-case analysis
Supply-risk deliverable proposed	Buyer buffer experiment and documented use limits

Delivered: Python pipeline, CSV forecasts, analysis reports and a model card.
The trading failures were already known at midsemester.

Data sources and preparation

Match the crop geography, the season and the information available at each decision

Weather and drought	NOAA nClimGrid county weather, NASA POWER points and US Drought Monitor drought coverage.
Yield and market data	OWID/FAO annual yield; USDA crop releases; NOAA CPC/GEFS forecasts; dated LSEG futures.
Preparation	Aggregate to growing seasons. Fit trends and scaling inside training folds. Use release times and next-open trades.
Key limitation	County history is revised and often late. Five-state equal weights do not represent national production weights.

No synthetic training data. Resampling estimates uncertainty; the buyer example simulates costs.

Baselines and model selection

The climate model must improve on a matched comparison with less information

Supply baseline

A linear yield trend, fitted only on earlier harvests. Compare errors on identical test years.

Supply model

Standardized Ridge, alpha = 1, fixed from the original study. July heat and peak drought explain yield deviations.

Forward evaluation

Start with ten training harvests. Refit trend and scaling each year. Report R^2 versus trend, MAE and interval coverage.

Trading models

Ridge, logistic regression and shallow trees. Select within earlier folds, then freeze. Compare with market-only and fixed rules.

Corn weather improves the yield estimate

Chronological evaluation with full-season county weather and drought

+0.748

R^2 versus trend across 16 harvests

+0.476

R^2 after removing 2012 and refitting

0.31 t/ha

MAE versus 0.58 for the trend baseline

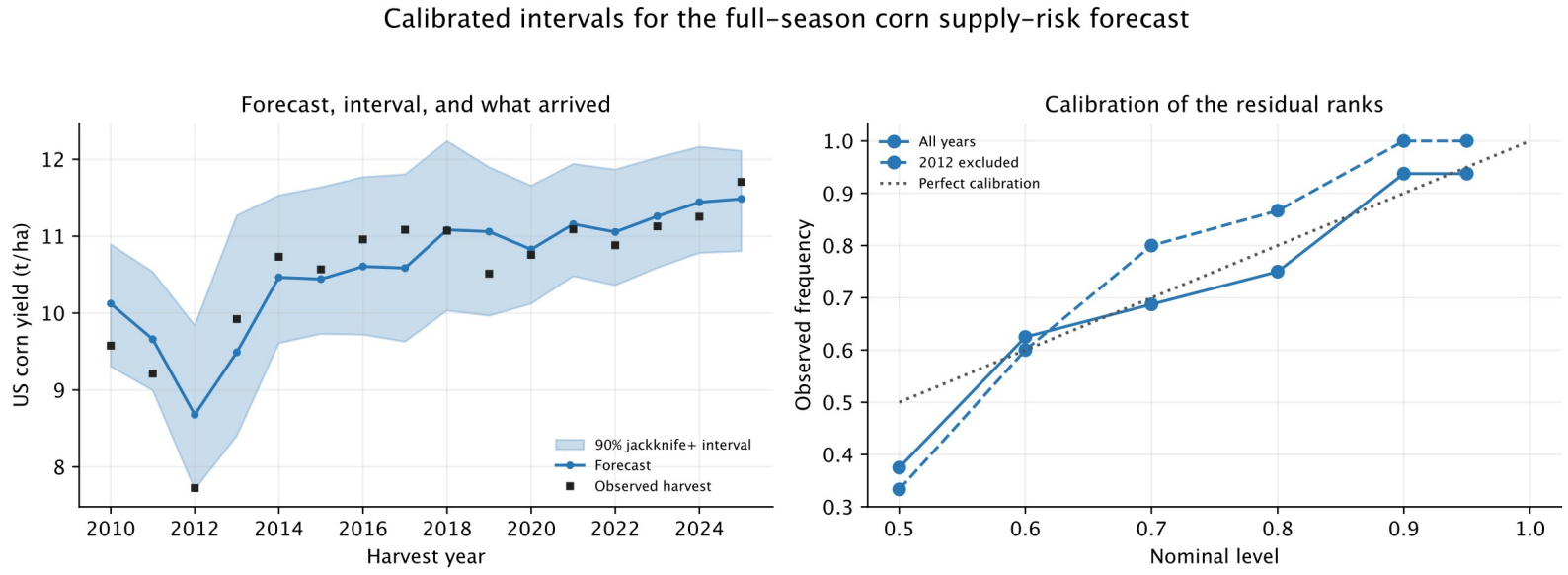
Improvement since midsemester: each test now uses only earlier training years, with a matched trend baseline.

The old leave-one-year-out R^2 uses a different denominator. These scores are not a before/after accuracy comparison.

Historical hindcast using revised inputs. No untouched yield holdout or timely county forecast is established.

The estimate now includes uncertainty

The interval must cover bad seasons as well as ordinary ones



Sixteen scored harvests: one season is worth 6.25 points of coverage, so these frequencies are coarse

16/16

historical harvests inside the nominal 90% interval

±8.7%

average half-width as a share of trend yield

0.018 t/ha

margin for the 2012 drought harvest

Observed coverage is not a future guarantee. The time series does not establish exchangeability.

The trading candidates failed validation

Success required positive net returns, stability and stronger results than matched controls

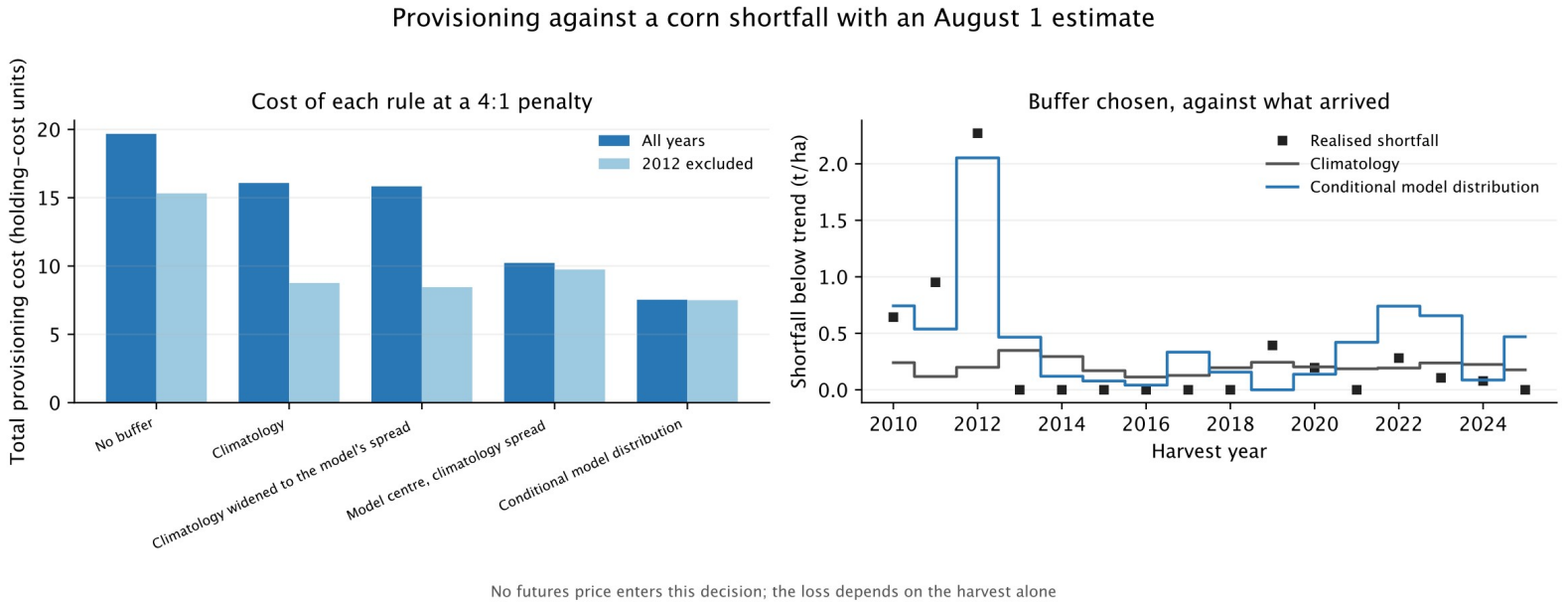
GEFS forecast revisions	**–7.3% net**	Sharpe −0.98 14 trades in the sealed 2026 holdout
USDA Crop Progress	**–52.7% net**	Sharpe −0.81 Positive in 3 of 9 development years

Seven USDA checks failed. Its 2026 holdout stayed sealed.
Frozen candidates were not retuned after failure.

New insight: weather adds +0.120 harvest R^2 beyond December futures prices, but only +0.001 without 2012. This suggests overlapping information.

A buyer can use the estimate to size a buffer

Worked example: an August 1 estimate from point weather and drought, at a 4:1 shortfall penalty



53%

lower simulated cost than a no-weather baseline

14%

estimated saving without 2012; uncertainty includes zero

Pilot first

analyst-reviewed risk reports for physical corn buyers

Hypothetical cost units, not realized dollars. Storage, basis and contracts are omitted; adoption is untested.

Risks, failure cases and responsible use

The model card limits the claims and the decisions this prototype can support

Geographic bias	Five states with equal weights. Test production weights and regional errors before extending to farms or other crops.
Uneven performance	Calm-season R^2 is −0.452 without 2012. Show results by season and retain the trend-only fallback.
Late or revised data	County inputs were timely in only 3 of 26 historical seasons. Archive future vintages and withhold stale outputs.
Decision safeguards	Human review and explicit intervals. No automated trading or procurement. Validate real costs and a new harvest first.

Real-world feasibility

The pipeline exists; production operation and commercial adoption remain future work

Implemented stack

Python, pandas, scikit-learn and Parquet. CLI ingestion produces reproducible forecasts and reports.

Proposed integration

Scheduled ingestion and freshness checks feed a versioned risk report for a procurement analyst.

Operational constraints

Data-release latency and revisions limit reliability. Public weather still needs maintenance; LSEG requires licensed access.

Scaling

Cache ingestion by region, monitor missing data, then validate each new crop. No production cost or speed benchmark yet.

OUTCOME AND REFLECTION

Public weather measures corn supply risk. The tested trading signals failed.

The lesson: useful physical information needs timely data and a validated decision before it creates value.

WHAT COMES NEXT

Freeze a new harvest test and compare with USDA forecasts.
Add production weights and pilot a buyer report with realistic costs.

Questions?

github.com/haydn-s/climate-futures-pricing